

FATIMA ABOUL

Software Developer | Computer Science Student

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SUMMARY

Motivated final-year Computer Science student with hands-on experience in software development, AI-powered applications, full-stack web development, and database-driven systems. Skilled in Python, JavaScript, React, Node.js, C#, and SQL, with a growing focus on modern AI integration. Eager to contribute to real-world software projects while continuing to grow as a developer.

EDUCATION

Bachelor of Science in Computer Science <i>DHA Suffa University, Karachi</i>	Oct 2023 – Present
Intermediate (Pre-Engineering) <i>TCF College, Qayyumabad, Karachi</i>	Aug 2021 – Jul 2023

EXPERIENCE

Tutor — FAHAD Tutors FTA - Tutored students of different age groups and skill levels. - Strengthened communication, writing, and interpersonal skills.	2024 – Present
Volunteer — Climate Change Initiative - Participated in environmental awareness and community engagement activities. - Developed a stronger sense of social responsibility and teamwork.	Jan 2023
Volunteer & Ambassador — The Citizens Foundation (TCF) - Proctored examinations with integrity in a fair, distraction-free environment. - Guided students and provided information about TCF programs and services.	2024 – Present

TECHNICAL SKILLS

Programming Languages: Python, C#, JavaScript, SQL
Web Technologies: HTML5, CSS3, React, Node.js, Express.js
AI / ML: Google Gemini API, RAG, Embeddings, Streamlit
Database & APIs: MySQL, REST APIs, JWT Authentication
Tools & Version Control: Git, GitHub, Visual Studio, VS Code, MS Word, PowerPoint

SOFT SKILLS

Communication, Teamwork, Leadership, Time Management, Problem Solving, Adaptability

PROJECTS

Flavr — Full-Stack Recipe Discovery App <i>React, Node.js, Express.js, MySQL</i> - Developed a full-stack recipe discovery platform with user authentication and recipe browsing. - Implemented favorites, weekly meal planning, image uploads, and dark mode. - Built an admin dashboard for recipe management and content control. - Integrated frontend, backend APIs, and MySQL database for efficient data management.
AI Agent <i>Python, Streamlit, Google Gemini API</i> - Built an AI agent using Gemini API and function calling to handle user requests. - Integrated tools for calculations, web search, date/time lookup, and live currency conversion. - Implemented multilingual responses with persistent chat history and a custom Streamlit interface. - Added user-friendly error handling for reliable interaction.
Retrieval-Augmented Generation (RAG) PDF Q&A Application <i>Python, Streamlit, Google Gemini API</i>

- Developed a RAG-based PDF question-answering application using Python, Streamlit, and Google Gemini API.
- Implemented text extraction, chunking, semantic embeddings, and cosine-similarity-based retrieval using pypdf and sentence-transformers.
- Deployed a live AI application that generates context-grounded answers from uploaded PDF documents.

Student Management System

C# Windows Forms, MySQL

- Developed a desktop application for managing student records.
- Implemented CRUD operations for adding, viewing, updating, searching, and deleting records.
- Added input validation to ensure accurate student data.
- Integrated MySQL database for reliable data storage and retrieval.

To-Do List App

HTML5, CSS3, JavaScript

- Developed a responsive task-management web application with an interactive user interface.
- Implemented task filtering, important-task marking, and localStorage persistence.
- Added calendar and task statistics views for better task organization.
- Integrated profile settings and dark mode for an enhanced user experience.